Robust Acoustic Positioning for Safety Applications in Underground Mining

Reimar Pfeil, *Member, IEEE*, Markus Pichler, *Member, IEEE*, Stefan Schuster, *Member, IEEE*, and Florian Hammer, *Member, IEEE*

***Abstract*— The surroundings of underground mining machines still constitute an unsafe area for miners due to bad visibility conditions. In this paper, we present a novel acoustic position estimation system for safety applications in such hazardous environments. Our system is based on a pulse compression technique and utilizes linear (LFM) and hyperbolic (HFM) frequency-modulated signals in the frequency range of 5–20 kHz. As the miners are moving, we incorporated a means of Doppler-shift compensation. This system not only provides the positions of moving targets but also their velocity and direction. In stationary measurements, we evaluated LFM signals at noisy conditions both in an indoor laboratory environment and in the vicinity of a mining machine. The movement of a miner has been emulated in dynamic laboratory measurements at constant speeds up to 1 m/s using both LFM and HFM signals. Our results show that the acoustic signal can be evaluated down to a low signal-to-noise ratio of −30 dB. The results of the dynamic measurements clearly demonstrate the insensitivity of the HFM signals to Doppler-shifts both with regard to the estimated position and estimated velocity.**

***Index Terms*— Collision avoidance, Doppler-shift compensation, hyperbolic frequency-modulated (HFM) chirp signals, pulse compression, robust acoustic position estimation, underground mining.**

I. Introduction

UNDERGROUND mining is an extremely harsh and dangerous business. According to the Mine Safety and Health Administration (MSHA)[1] in the United States, approximately 22% of all mining-related fatalities occurred due to visibility issues of machinery in coal mines. A typical mining accident situation is shown in Fig. 1. Such a situation should be prevented by all cost by an emergency stop if a miner gets too close to an operating machine. Clearly, this requires the knowledge of the location of the miner relative to the machine which has to be determined:

1) fast (miner is moving);
2) accurately (the hazardous zones should be restricted to certain areas in the vicinity of the machine);
3) reliably (low false alarm rate is mandatory).

Moreover, the desired positioning system should also be mechanically robust and not require an elaborate installation, e.g., devices surrounding the measurement field. The points 1 and 2 of the list can be relaxed since the tolerable moving speed in a mine is approximately 1–5 m/s and the accurate positioning suffices within a 50-m radial distance from the mining machine. In the vicinity of the machine, where multipath reflections become an issue, the use of a large signal bandwidth allows for the discrimination of the line-of-sight signal and unwanted reflections [1].



Fig. 1. Example of a possible mining accident in an underground coal mine. The miner (left) tries to operate the machine although he can get easily squeezed between the machine and the mine wall (right). In this case, the proposed system would protect the miner against accidental maloperation or operation of the machine by another person.

A. Concepts for Accident Avoidance

Positioning in underground mines has been covered in [2]–[6] and most of them use received signal strength indication (RSSI), laser-based range finder, or inertial navigation systems. Recently, Bedford and Kennedy [7] have analyzed the applicability of ZigBee (IEEE 802.15.4) devices for time-of-flight (TOF) measurements. It should be noted that numerous other radio frequency (RF)-based local positioning systems [8]–[11] are currently available on the market which are also using TOF measurements and promise centimeter-level accuracy. However, the performance of these systems deteriorates drastically in dense multipath (indoor)

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[1] http://www.msha.gov

[2] http://www.imr.rwth-aachen.de/index.php/featureface-191.html

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scenarios with many (metal) reflectors. Note that, in general, an underground mine is equipped with metal shaft lining and cable trays. Hence, even ultrawideband systems [11] are struggling with the severe multipath situation which often leads to unacceptable outliers in the position estimates. Hence, a new approach for the positioning problems becomes inevitable, which is presented in the following section.

For more details on wireless position estimation techniques and systems, we refer the reader to [12].

B. Acoustic Position Estimation

A way of dealing with the harsh environment of a mine is to use a position estimation method that employs acoustic sound waves. Acoustic position estimation can be based on ultrasound or audible sound. In the ultrasound range, the Active Bat [13] and Cricket [14] systems use TOF measurements to estimate the position. While in the Active Bat system, the mobile entity emits the signal to a fixed set of receivers, in the Cricket system, the ultrasound signal and a RF trigger are emitted by beacons fixed on the surrounding walls and the mobile entity calculates its position on board. Similar to the Active Bat system, the acoustic self-calibrating system for indoor smartphone tracking [15] emits high-pitched frequency-modulated signals from a smartphone to fixed receiver stations and calculates the position of the smartphone based on time difference of arrival measurements.

In simultaneous range measurements, the channels can be distinguished by the application of coding to the individual measurement signals. Direct sequence spread-spectrum techniques are used in the DOLPHIN (Distributed Object Locating system for PHysical-space INternetworking, [16]) and in the system described in [17]. Both of these systems use Gold codes. A 32-bit Golay code is utilized in the 3D-LOCUS [18] system.

The position of an object may be determined by means of source sound localization employing microphone arrays [19], [20]. As the acoustic environment of a mine consists of a considerable amount of noise, and as the emission of loud sound signals from the miners side is not feasible due to limited battery capacity, we do not further elaborate on this topic. Regarding underwater localization techniques, we refer the interested reader to [21].

This paper is structured as follows. In Section III, the signal processing is presented which is used to evaluate the range measurements. Furthermore, a statistical analysis of the range estimation process is given in Section III-A which is used to select the parameters to meet the system requirements in Section III-B. In Section III-C, the Doppler influence is discussed and two methods are presented to overcome the Doppler-related issues. In Section III-D, a short overview of the used position estimation methods is given. Section IV is centered around the measurements that were acquired at various stages in this project. First, only static measurements have been performed which are summarized in Section IV-A. Second, the Doppler-shift has been analyzed more precisely and the according measurements results are conducted in Section IV-B. A summary of all achievements is given in Section V.

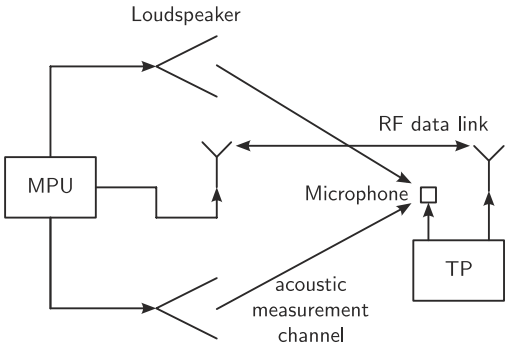


Fig. 2. Schematic of the acoustic positioning concept. The master processing unit transmits an RF trigger to the transponder which then starts the acoustic recording. At the same time, the master processing unit emits the acoustic signals through the LSs which are received with a TOF delay at the transponder. The recorded acoustic signal can either be evaluated directly at the transponder or transmitted back to the master processing unit through the RF data link.

II. Methodology for Acoustic Positioning

The basic principle of our approach is based on the well-known thunder and lightning concept. The distance of a person to a thunderstorm can be roughly estimated by the time between the visible lighting and the heard thunder. Since the propagation velocity of an electromagnetic (EM) wave ($c_{\mathrm{EM}} \approx 3 \cdot 10^8$ m/s) is much larger than the propagation velocity of sound ($c_s \approx 340$ m/s), the TOF delay of the EM wave becomes negligible compared with the acoustic wave. Thus, two spatially separated devices can be synchronized by EM waves while their acoustic emission and reception is measured. This allows to determine the absolute ranges between the moving target (MT) and the mining machine. A schematic of this approach is shown in Fig. 2. Two different loudspeaker (LS) locations and one constraint are necessary to determine the 2-D position of the MT. For the application in a mine, the LSs are assumed to be mounted on the mining machine heading toward the access area. Hence, the constraint is simply given by the restriction to the positive half-plane in front of the LS array. The master processing unit coordinates the transmission of the acoustic signals and the RF trigger to the transponder mounted at the MT. The recorded signal at the transponder can either be directly returned to the master processing unit using a wireless communication link or evaluated in the transponder using a digital signal processor (DSP).

A. Acoustic Measurement Channel

While the number of the passive MTs can be theoretically unlimited, the number of the active LSs is limited by the access to the acoustic channel. In this contribution, we focus on the usage of code, frequency and space division multiple access techniques to deal with this limitation. The latter is inherently used since the LSs and the microphones are supposed to have directional characteristics. Note that an important premise for precise and accurate positioning is the restriction to line-of-sight paths only and to discard all nonline-of-sight corrupted measurements [22]. Hence, it cannot be emphasized enough that the best approach to achieve robust position estimates is a neat signal processing.

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B. RF Transmission Channel

The RF channel is used for two purposes. First, it triggers the transponder to start the recording and second, it is used to transmit the recorded data back to the master processing unit. The amount of the transmission data is dependent on the tasks performed at the transponder. The most simple realization would be a transponder streaming the acoustic signal back to the master processing unit via a communication link in the very or ultrahigh-frequency band [23, p. 1.15]. This approach would have the significant benefit that the acoustic signal can be sampled with a high sampling frequency and processed in a computational exhaustive way at the master processing unit. Another method would be the processing of the acoustic signal with a DSP at the transponder and only to transmit the estimated ranges or position back to the master processing unit. For the development of a prototype system, the latter approach is pursued. Note that for the test measurements presented in this paper that were performed at an early development stage, a wired connection used instead of the RF link. Here, the signal processing was performed on a PC.

C. System Integration

The acoustic positioning system should be integrated in a collision avoidance system [24], which operates in the whole mine. To keep the sonic emissions to a minimum level, the positioning determination process is only started if the MT gets within the range of the acoustic system. For this purpose, a coarse estimate of the radial distance of the MT to the mining machine is sufficient for which an RSSI-based system can be used. Furthermore, personnel operating at a safe driver's cabin should not trigger an acoustic estimation process.

III. Acoustic Signal Processing

A crucial point of this approach is that the acoustic signal must be robust against disturbing noise, e.g., by drilling machines, and allow precise and accurate determination of the TOF. For this purpose, we implemented a pulse compression technique with a chirp transmission signal [23, Ch. 8], [25]. A basic pulse compression radar uses a matched filter to achieve a maximum signal-to-noise ratio (SNR) and thus assures robustness against the disturbing noise. In a practical digital system, the matched filter is implemented as a correlation. In a first approach, a linear frequency modulation (LFM) chirp signal was used which is given by

$$s_{\mathrm{LFM}}(t) = A\cos\left(2\pi\left(f_0 + \frac{\alpha}{2}t\right)t + \varphi_0\right), \quad (0 \le t < T) \quad (1)$$

where $A > 0$ is the amplitude, T is the pulsewidth, φ_0 is the initial phase, f_0 is the frequency, and LFM is the slope

$$\alpha = \frac{B}{T} \quad (2)$$

where $B = f_1 - f_0$ denotes the bandwidth and f_1, the stop frequency of the chirp. This signal is robust against interference and can be easily generated. In the absence of multipath reflections, the received signal at the MT can thus be modeled as the sum of the time-delayed variants of (1) given by

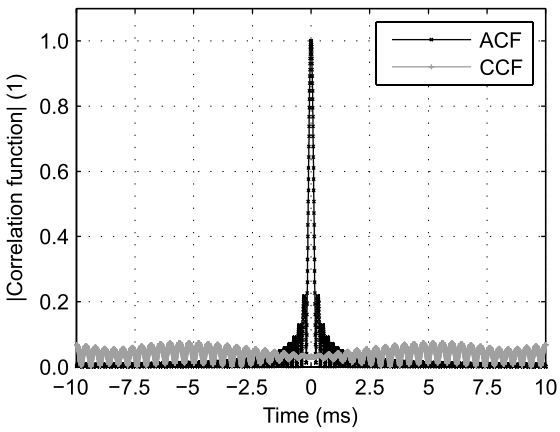


Fig. 3. Absolute of auto- and cross-correlation function (ACF/CCF) of an up-chirp ($f_0 = 5$ kHz, $f_1 = 10$ kHz) and down-chirp ($f_0 = 10$ kHz, $f_1 = 5$ kHz) with $T = 100$ ms and $A = 1$. For the time discretization a sampling rate of $f_s = 192$ kHz was used.

$$s^r_{\mathrm{LFM}}(t) = \sum_{p=1}^{P} \kappa_p s_{\mathrm{LFM}}(t - \tau_p) \quad (3)$$

where τ_p is the TOF from the pth LS to the microphone, κ_p is the free-space attenuation loss, and P is the total number of LSs in the array. Equation (1) can either be an up-chirp ($f_1 > f_0 \rightarrow \alpha > 0$) or a down-chirp ($f_1 < f_0 \rightarrow \alpha < 0$) signal. As shown in Fig. 3, the correlation of an up-chirp with a down-chirp signal results in a low cross-correlation compared with the autocorrelation even when the chirps use the same frequency band. This property is exploited to separate the signals from the two LSs (Fig. 2).

For the evaluation, the recorded signal $y(t)$ is sampled using a sampling frequency $f_s > \max(2f_0, 2f_1)$ yielding

$$y[n] = [\, y[0] \quad y[1] \quad \ldots \quad y[N_r]\,] \quad (4)$$

where

$$N_r = \lceil (T + \tau_{\max}) f_s \rceil \quad (5)$$

with the maximum expected TOF $\tau_{\max}$ and $\lceil \circ \rceil$ denoting the ceil operator. The time-discrete version of (1) is given by

$$s_{\mathrm{LFM}}[n] = A\cos\left(2\pi\left(\psi_0 + \frac{\beta}{2}n\right)n + \varphi_0\right), \quad (0 \le n \le N) \quad (6)$$

with $\psi_0 = f_0/f_s$, $\beta = \alpha/f_s^2$, and N given by

$$N = \lceil T f_s \rceil. \quad (7)$$

The correlation can efficiently be computed in the frequency domain if zero padding is applied so that the multiplied vectors have the same length of at least $N_r + N - 1$ to avoid circular correlation effects, resulting in

$$s_c[n] = \mathrm{IDFT}\{\mathrm{DFT}\{\mathbf{y}\} \cdot \mathrm{DFT}\{\mathbf{s}_{\mathrm{LFM}}\}^*\} \quad (8)$$

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with $\circ^*$ denoting the complex conjugate, and DFT denoting the discrete Fourier transform defined as

$$X[k] = \mathrm{DFT}\{x\}[k] \triangleq \sum_{n=0}^{N-1} x[n] \exp\left(-\mathrm{j}2\pi \frac{nk}{N}\right). \quad (9)$$

The inverse DFT (IDFT) in (8) has the same expression as (9) but with a positive sign in the exponent and is given by

$$x[n] = \mathrm{IDFT}\{X\}[n] \triangleq \frac{1}{N} \sum_{k=0}^{N-1} X[k] \exp\left(\mathrm{j}2\pi \frac{nk}{N}\right). \quad (10)$$

Both can be efficiently evaluated using the fast Fourier transform algorithm [26]. Note that if before applying the IDFT, the frequency-domain signal is padded with zeros at half the sampling frequency to Z times its length, an interpolated variant of (8) is obtained [27]. Usually, a zero-padding factor of 4 is enough for the systematic interpolation errors to be well below the measurement errors. The TOFs τ_p can then be estimated by

$$\hat{\tau}_p = \frac{T_s}{Z} \arg\max |s_c[n]| \quad (11)$$

with the sampling period $T_s = 1/f_s$ and $\hat{\circ}$ denoting the estimated value. Note that (11) underlines the importance of a zero padding in frequency domain, because otherwise the estimation error can be $\pm T_s$. Aside the zero-padding approach presented in [27], an inverse chirp z-transform algorithm (CTA) [28] can also be used to achieve a better time-domain resolution.

A. Statistical Analysis

The statistical analysis of the pulse-based range estimation is already covered in [29, pp. 53–56] and [30]. Hence, only the main results are restated and adapted for the given signal. The Cramér–Rao lower bound (CRLB) of the estimate (11) is given by

$$\mathrm{var}\{\hat{\tau}_p\} \geq \frac{1}{\bar{B}^2 \frac{\eta}{T_s} E} \quad (12)$$

with the mean square bandwidth of the transmission signal $s(t)$

$$\bar{B}^2 = \frac{\int_0^T \left(\frac{ds(t)}{dt}\right)^2 \mathrm{d}t}{\int_0^T s^2(t) dt} \quad (13)$$

the normalized energy

$$E = \int_0^T s^2(t) dt \quad (14)$$

and the SNR definition

$$\eta = \frac{A^2}{2\sigma^2} \quad (15)$$

where σ^2 denotes the noise variance. The CRLB for the TOF can be converted to range variance by

$$\mathrm{var}\{\hat{r}_p\} \geq \frac{c_s^2}{\bar{B}^2 \frac{\eta}{T_s} E} \quad (16)$$

since $\mathrm{var}\{ax\} = a^2 \mathrm{var}\{x\}$ with an arbitrary constant a and a random variable x.

B. Parameter Selection

With the knowledge of the CRLB, the choices for the duration T and the bandwidth B can be evaluated for their influence on the achieved performance. Clearly, the start frequency f_0 has no influence on the performance and therefore should be chosen based on system (i.e., according to the transfer function of LS and microphone) and measurement (i.e., frequency band with less disturbing noise) limitations.

Another important figure of merit is the time delay and range resolution [23, pp. 8.4–8.5], which is given by

$$\Delta r_{3.9} = \frac{c_s}{B} \quad (17)$$

where $\Delta r_{3.9}$ denotes the range resolution for the 3.9-dB mainlobe width. Equation (17) describes the minimal distance between two signals to be resolvable as single targets in a bistatic configuration. This parameter also defines the minimal distance between a reflected nonline-of-sight and line-of-sight signal to be separable. As in air approximately $c_s \propto \sqrt{T}$, and the variation in temperature T is in the range of at most several tens of K, the temperature effect on the achievable range resolution can be considered negligible.

C. Doppler Compensation

A major issue for all acoustic systems is the large influence of the Doppler effect [34]. For the given case that the target is moving at a constant velocity during signal transmission, and the emitters are stationary,[3] the received frequency f_r is shifted according to

$$f_r = f_e \left(1 - \frac{v_{\mathrm{MT}}}{c_s}\right) \quad (18)$$

where v_{MT} is the radial velocity of the MT and f_e is the emitted frequency [35, p. 415]. For a movement toward the receiver, $v_{\mathrm{MT}} < 0$ and $v_{\mathrm{MT}} > 0$ otherwise.

1) Doppler Compensation of an LFM Signal: The consideration of the Doppler influence yields that (3) has to be replaced by

$$s_{\mathrm{LFM}}^{r,d}(t) = \sum_{p=1}^{P} \kappa_p s_{\mathrm{LFM}}^{d}(t - \tau_p, v_{\mathrm{MT},p}) \quad (19)$$

with the radial velocity $v_{\mathrm{MT},p}$ of the MT toward the pth LS

$$s_{\mathrm{LFM}}^{d}(t, v_{\mathrm{MT}}) = A \cos\left(\phi_{\mathrm{LFM}}^{d}(t, v_{\mathrm{MT}})\right), \quad (0 \leq t < T) \quad (20)$$

and the abbreviation for the phase argument

$$\phi_{\mathrm{LFM}}^{d}(t, v_{\mathrm{MT}}) = 2\pi \left(f_0 + \frac{\alpha}{2} t\right) \left(1 - \frac{v_{\mathrm{MT}}}{c_s}\right) t + \varphi_0. \quad (21)$$

Note that the Doppler influence alters the start frequency f_0 and the slope rate α. This creates the critical problem that the received LFM signal gets more dissimilar to (1) for an increasing T and v_{MT}. Thus, the correlation result degrades, which is demonstrated in Fig. 4. Moreover, no unique maximum value can be determined. This problem is also known as the range-Doppler coupling [23, p. 8.4], which can be

[3]The slow movement of the mining machine can safely be considered negligible.

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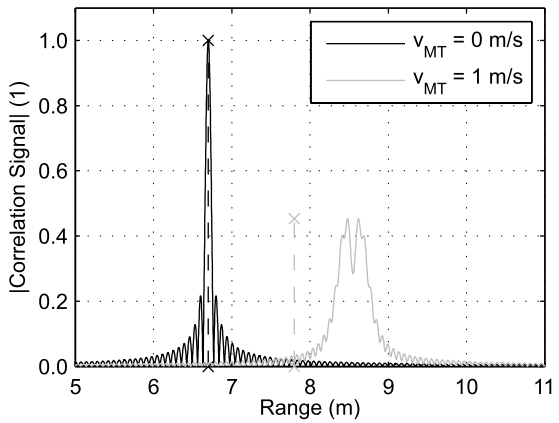


Fig. 4. Absolute value of correlation result of an LFM chirp signal with $f_0 = B = 5$ kHz, $T = 500$ ms sampled at $f_s = 192$ kHz, and a zero-padding of $Z = 2^{18}$. The simulated distances are $r_1 = 6.7$ m for the black result and $r_2 = 7.8$ m for the gray result (both marked by vertical lines with $\times$ markers).

solved using Costa arrays [36] instead of an LFM chirp or a range-Doppler map [37], [38, p. 281]. Other approaches are found in [39].

However, these solutions have certain drawbacks: while Costa arrays are only known to a limited number of frequency steps and their evaluation is computationally quite demanding, the range-Doppler map approach has a certain limit for the maximum velocity, where the v_{MT} estimates become ambiguous. This boundary is inverse proportional to the ratio between propagation velocity and the bandwidth, and thus the maximum velocity is very low for acoustic signals.

Hence, we propose the application of the discrete polynomial-phase transform (DPT) [40] to estimate the three polynomial coefficients of (21) and then to derive estimates for the TOF and velocity. The details of this procedure are summarized in Appendix A. The final velocity estimate (39) is the scalar projection of the velocity vector of the target on the directional vector from the LS to the MT. The conversion of (39) to a 2-D velocity vector is given in Appendix B.

As described in [40], the DPT is a cyclic algorithm which decomposes the polynomial-phase signal starting with highest order of the polynomial, in our case, the second order. Within each decomposition step the DPT uses the Schuster periodogram [41, pp. 966–971] to estimate the according polynomial coefficient. To prevent a large error propagation between the two estimates, a high grid resolution in the DFT-based estimation process is of great importance [42]. Hence, we suggest the application of the CTA and a spectral peak location algorithm [33, pp. 523–525] which is also known as a parabola estimation (PE) [43].

2) Doppler Compensation Using Chirps With Hyperbolic Frequency Modulation: In [44], it is shown that hyperbolic-frequency modulated (HFM) chirps have an optimal modulation property or Doppler invariance which was already derived in [45] and [46]. The HFM chirp is given by

$$s_{\mathrm{HFM}}(t) = A\cos(\phi_{\mathrm{HFM}}(t)), \quad (0 \le t < T) \tag{22}$$

with the phase argument

$$\phi_{\mathrm{HFM}}(t) = \frac{2\pi}{b}\ln(1 + bf_0 t) \tag{23}$$

where the modulation parameter b is given by

$$b = \frac{f_0 - f_1}{f_0 f_1 T}. \tag{24}$$

As for the LFM chirp, $f_0 < f_1 \rightarrow b < 0$ is an up-chirp, while $f_0 > f_1 \rightarrow b > 0$ denotes a down-chirp. Assuming a TOF τ_p and a velocity $v_{\mathrm{MT},p}$ yields for (23)

$$\phi^d_{\mathrm{HFM}}(t - \tau_p, v_{\mathrm{MT},p}) = \frac{2\pi}{b}\ln(1 + bf_0\nu_p(t - \tau_p)) \tag{25}$$

where the abbreviation (33) was used. As shown in [44], the instantaneous frequency is expressed as

$$\begin{aligned} f_{\mathrm{HFM}}(t - \tau_p, v_{\mathrm{MT},p}) &= \frac{1}{2\pi}\frac{\partial}{\partial t}\left[\phi^d_{\mathrm{HFM}}(t - \tau_p, v_{\mathrm{MT},p})\right] \\ &= \frac{\nu_p f_0}{1 + bf_0\nu_p(t - \tau_p)} \end{aligned}$$

which can be equivalently written as

$$f_{\mathrm{HFM}}(t - \tau_p, v_{\mathrm{MT},p}) = \frac{f_0}{1 + bf_0(t - \tau_p - \tau_{d,p})} \tag{26}$$

with

$$\tau_{d,p} = \frac{\phi_p}{bf_0} \tag{27}$$

and

$$\phi_p = 1 - \frac{1}{\nu_p}. \tag{28}$$

Equation (26) is quite remarkable, since the Doppler contribution can be modeled as a simple time-delay $\tau_{d,p}$. This can also be validated by a repetition of the simulation from Fig. 4 shown in Fig. 5. Clearly, the Doppler shift has no effect on the shape of the correlation peak anymore and is only shifted by $\tau_{d,p} + \tau_p$. This can be exploited in the following way: by emitting an HFM up-chirp and down-chirp from *each* LS, the TOF τ_p and velocity $v_{\mathrm{MT},p}$ can be estimated. The details of this procedure are summarized in Appendix C. Note that the modulation parameters of the two chirps at each LS do not have to have an opposing sign, i.e., two up-chirps or down-chirps can also be used. This originates from another beneficial property of the HFM that chirp signals can overlap in their frequency band without creating an additional spectral peak which is shown in Fig. 6.

The plots in Fig. 6 are deliberately zoomed out to illustrate the cross-correlation effects starting at $|r| \gtrapprox 17$ m. Note that the contribution from the opposite chirp in Fig. 6(a) is a noise shaped, while in Fig. 6(b) is a rectangular shaped signal. Hence, it is beneficial to mix both methods for the transmission which we used by transmitting an up-chirp and down-chirp for each LS and separating each LS in the frequency domain.

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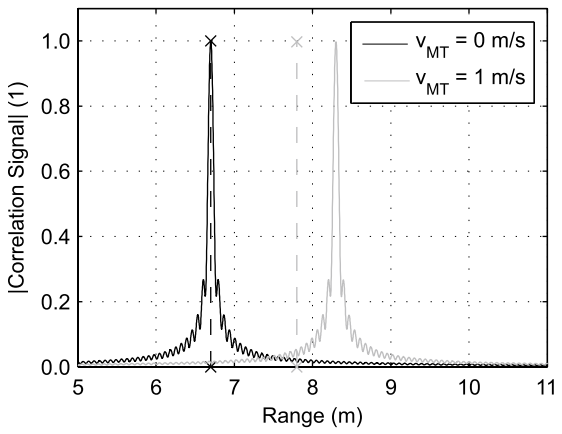
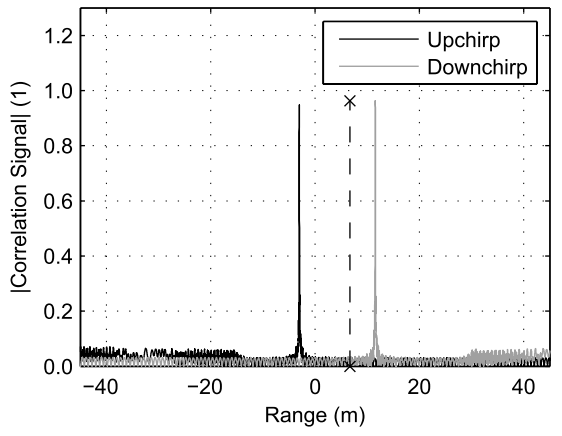
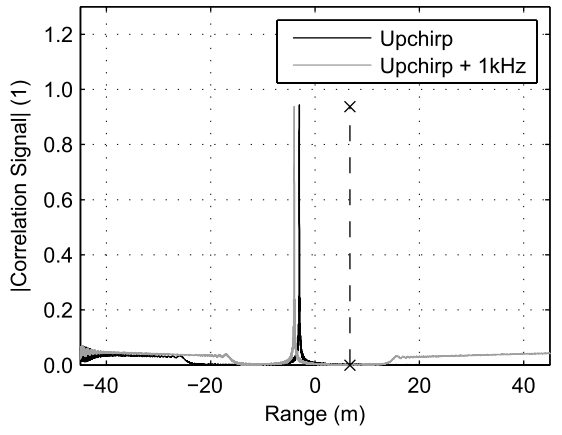


Fig. 5. Repetition of the simulation from Fig. 4 with the same signal parameters but with an HFM instead of an LFM chirp.



(a)



(b)

Fig. 6. Signal separation at each LS using (a) an up-chirp and a down-chirp ($f_{1,0} = 5$ kHz, $f_{2,0} = 10$ kHz) or (b) two up-chirps with different start frequencies ($f_{1,0} = 5$ kHz, $f_{2,0} = 6$ kHz). For both cases, the chirp parameters $B = 5$ kHz and $T = 500$ ms sampled at $f_s = 192$ kHz and a zero padding of $Z = 2^{19}$ were used. The simulated distance was $r_1 = 6.7$ m and the velocity $v_1 = 10$ m/s.

D. Position Estimation

As previously stated, the basic concept is tailored to a 2-D position estimation which suffices for the task of the estimation of the MTs location around the machine. This leads to a simple trilateralization which is derived in [47]. In essence, the measurement from every LS yields a range to the mobile tag of which the position is calculated by intersecting circles with their centers at the LS positions. If more LSs are used, and thus more ranges are measured, a Gauss–Newton iteration [29, pp. 258–260] or a noniterative approach [48, pp. 162–163] can be applied. However, to achieve a robust position estimate it is preferable to first acquire position estimates from a minimal set of measurements and then fuse the results using a least median of squares algorithm [49].

IV. Measurement Results

To verify the previous simulation results, numerous measurements have been performed at various points in the project. Two LSs were setup and the actual positions of their centerpoints were measured using a laser distance meter. For all following measurements, a linear 16-bit quantization of the audio signal is used. The definition of the number of recorded samples given in (5) is depending on the maximum range of the acoustic system. As previously stated, the acoustic positioning system should only be used to locate the MT in the vicinity of the mining machine, and thus, a maximum range of 50 m is sufficient. Thus

$$\tau_{\max} = 50 \text{ m}/c_s \approx 147 \text{ ms}. \tag{29}$$

A. Static Measurements

In the first stage of this project, it was imperative to test if acoustic positioning is truly possible at noisy conditions inside a mine and close to mining machinery. For the first tests only LFM chirps were used.

1) Indoor Lab Measurements: A basic acoustic measurement system was assembled using off-the-shelf audio hardware as summarized in Table I. Synchronous audio signal generation and capture was performed using the Edirol UA-101 audio interface. The signal parameters for the first tests are given in Table II. Note that the down-chirp for the second LS is realized by exchanging f_0 with f_1. If more than two LSs were used, the individual signals could be differentiated using different (instantaneous) frequencies for every speaker. Since the RF transmission of the acoustic signal from the microphone back to interface would require a substantial development effort, we simply used a cable to directly connect the microphone to the audio interface. To test the location estimation performance, the microphone was mounted on a 2-D positioning system, which allows for positioning with centimeter accuracy. A photo of the setup is shown in Fig. 7 and the position estimates compared with the reference positions in Fig. 8. Furthermore, the cumulative distribution probability (CDP) of the location estimation error [48, p. 163] is shown in Fig. 9, which is defined as the L_2-norm distance between the reference and the estimated one, or

$$\|\hat{\mathbf{p}} - \mathbf{p}\|_2 = \sqrt{(\hat{p}_x - p_x)^2 + (\hat{p}_y - p_y)^2} \tag{30}$$

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TABLE I
AUDIO HARDWARE USED FOR THE FIRST TEST OF THE ACOUSTIC POSITIONING SYSTEM

Device	Product title
Loudspeaker	Visaton W100S - 4Ω
Audio interface	Edirol UA-101
Microphone	Unbranded dynamic Mic. with cardoid polar pattern sensitivity $-59\,\mathrm{dB} \pm 3\,\mathrm{dB}$ frequency range $80\,\mathrm{Hz} - 12\,\mathrm{kHz}$ output impedance $500\,\Omega$
Amplifier	Raveland XA1000

TABLE II
SIGNAL PARAMETERS FOR THE ACOUSTIC SIGNALS FOR THE FIRST LAB TEST

Parameter	Value
Sampling frequency f_s	44.1 kHz
Duration T	100 ms
Start frequency f_0	500 Hz
Stop frequency f_1	10.5 kHz

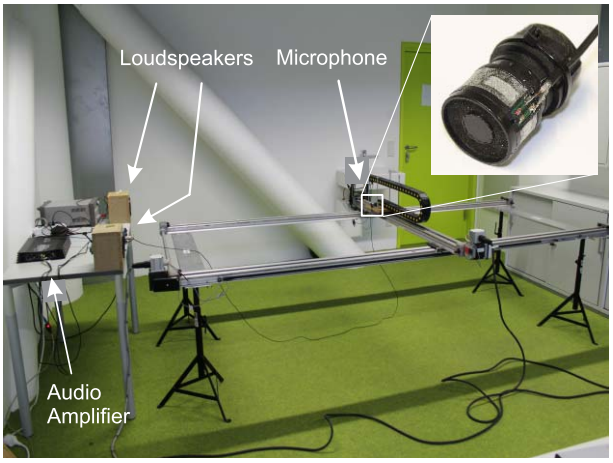


Fig. 7. Photo of first lab indoor measurements with acoustic positioning system.

where $\mathbf{p} = [\,p_x\ p_y\,]^T$ denotes the reference and $\hat{\mathbf{p}} = [\,\hat{p}_x\ \hat{p}_y\,]^T$, the estimated 2-D position. For the position estimation, a simple trilateralization approach is used that computes the intersection between the two range circles and only returns the result for the positive x-dimension.

Note that the accuracy of all measurements is below 3 cm. Encouraged by these good results, we repeated the measurements with typical mining tools (e.g., drilling machine and angle grinder) running in parallel. However, the position estimates became much worse which lead to the following adoptions.

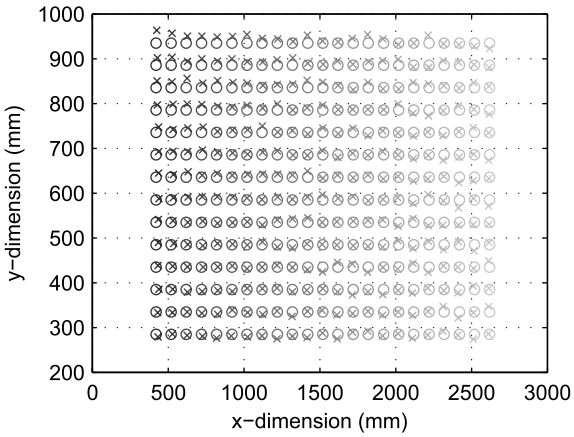


Fig. 8. Comparison of reference positions (gray circles) with estimates (gray crosses) of the acoustic position estimations setup from Fig. 7. The 2-D measurement axes were moved 23 times in the x-dimension with an increment of $\Delta x = 10$ cm and 14 times in the y-dimension with an increment of $\Delta y = 5$ cm yielding a measurement area of 230 × 70 cm and 322 measurements.

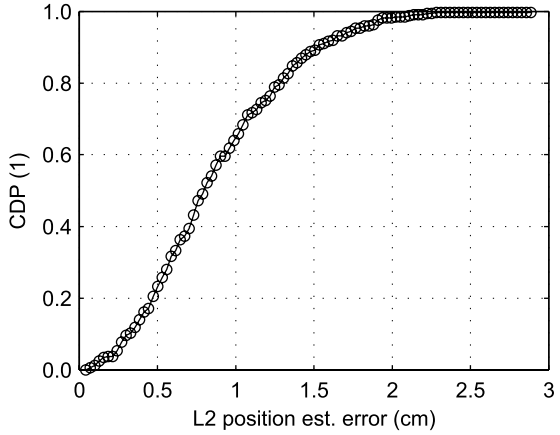


Fig. 9. CDP of the position estimation error of the indoor lab measurements (Fig. 7).

1) Since all tools have their most acoustic emissions below 5 kHz, we used $f_0 = 5$ kHz for the further measurements.
2) The duration of the chirp was increased to $T = 500$ ms.

The repeated noise measurement results are very similar to those in Fig. 8 and are not shown here. The results from using additional LSs as noise sources emitting a measured noise-level corresponding to the appropriate SNR are shown in Fig. 10, which plots the CRLB corresponding to the chosen chirp duration.

2) Test at Mining Machine: The basic acoustic system was also tested directly at a mining machine in an industrial environment. A test in a coal mine would be barely possible since it would require all components to be ATEX (Atmosphères Explosibles) certified [50]. A photo and schematic of the setup are shown in Fig. 11. Note that the machine was encased by a metal shaft lining to simulate a mine-like environment. In contrast to the previous measurements, four JBL Control-AV25 LSs are now used, which provide a frequency range from 70 Hz to 23 kHz. Moreover, the

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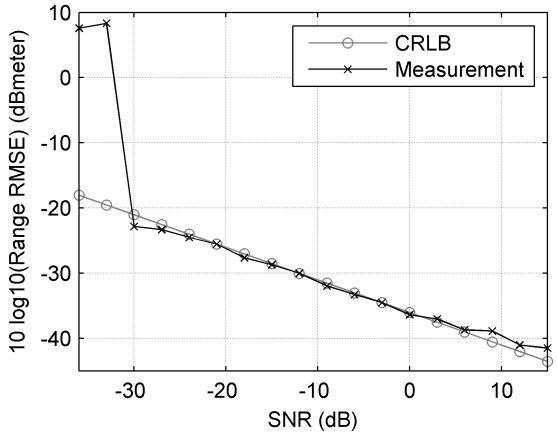
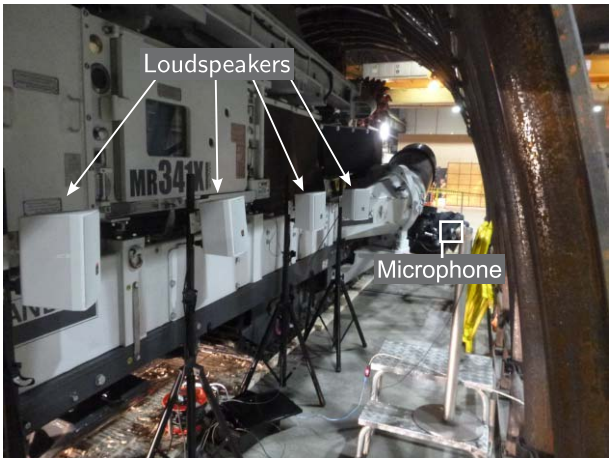
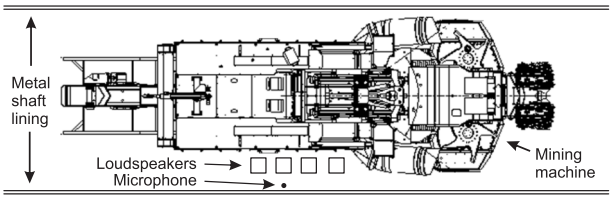


Fig. 10. Comparison of CRLB with the measurement result. The chirp parameters were $T = 500$ ms, $f_0 = B = 5$ kHz. For each SNR level, 200 measurement trials were recorded at a sampling rate of $f_s = 44.1$ kHz. To achieve a sufficient grid resolution, an equivalent zero padding of $Z = 2^{20}$ and a PE algorithm was used.



(a)



(b)

Fig. 11. (a) Photo and (b) top-viewed schematic of the measurement setup at a mining machine.

application of four LSs creates redundancy if one or two LS measurements fail. For the position estimation a Gauss–Newton approach [29, pp. 258–60] is applied. In Fig. 12, the position estimation results are summarized for various test cases I–V which conditions are explained in Table III. For each test case 30 measurements were performed but unfortunately all position estimates are biased to a certain degree. Since a bias of approximately 19 cm is also present when the machine was OFF, it is likely to be an error in the reference position of

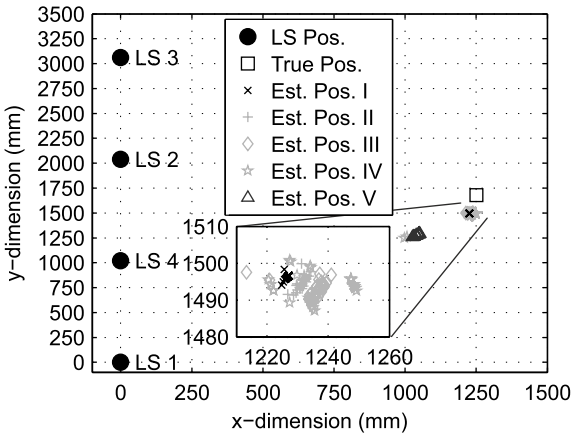


Fig. 12. Position estimation results of test at a mining machine. The region of interest is zoomed in for clarity.

the microphone or the setup of the LSs. Still, the measurements allow a qualitative comparison of the given test cases.

1) The position estimates between test cases I and II are almost identical. Hence, it can be concluded that the machine noise has no effect on the position estimation for the chosen parameters.
2) Even if the chirp duration is lowered by a third, the position estimates still remain around the reference position.
3) By additionally lowering the amplitude to a tenth, some estimates become outliers while the rest are still around the position of the previous measurements. This behavior originates from false range measurements of one or more LSs and marks the threshold level of the position estimation process, i.e., the signal falling below the detection threshold.
4) The last test case was performed with only a little lower amplitude than test case IV, but now *all* position estimates are outliers. This verifies the threshold-level assumption of test case IV.

In addition to the position estimation measurements, the sound pressure levels (SPLs) were measured using an AL1 Acoustilyzer from NTi Audio which conforms to the IEC 61672 standard. The SPL is given by

$$\mathrm{SPL} = 20 \log_{10} \left(\frac{p_{\mathrm{rms}}}{p_{\mathrm{ref}}} \right) \tag{31}$$

with the time-mean-square pressure of sound p_{rms} and the reference sound pressure $p_{\mathrm{ref}} = 20\ \mu$Pa. The measurement result plotted against frequency is shown in Fig. 13. Clearly, the most significant noise signals are concentrated below 5 kHz, and thus, emitting the chirp signals in the higher frequency domain is indeed advantageous.

B. Dynamic Measurements

As shown in Fig. 4, the positioning of an MT has drastic consequences on the correlation result. To verify the simulation results, a single linear axis was used which provides accurate driving speeds. The selected velocities are summarized in Table V with their according markers and colors used for all

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TABLE III
PARAMETERS OF TEST CASES USED DURING THE MEASUREMENTS FROM FIG. 12

Test Case	Amplitude	Duration	Machine State
I	100%	300 ms	off
II	100%	300 ms	on
III	100%	100 ms	on
IV	10%	100 ms	on
V	7.5%	100 ms	on

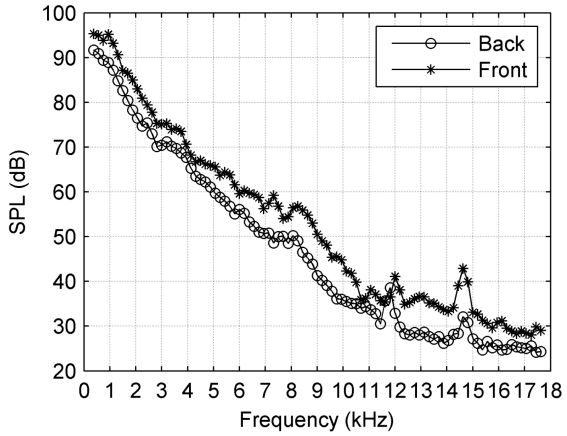


Fig. 13. Recorded SPL plotted against frequency in the back and front of the mining machine.

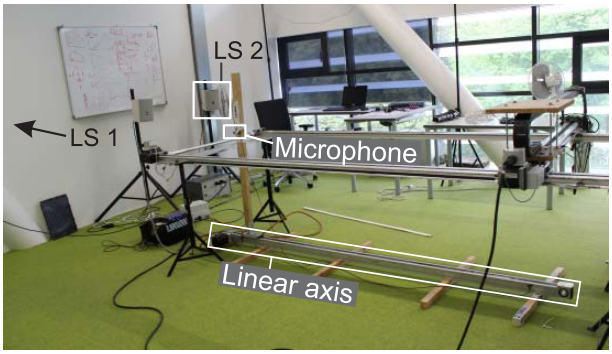


Fig. 14. Photo of the setup of the second lab measurement series. The microphone was mounted on a wooden beam which itself was moved at given velocities by the linear axis.

TABLE IV
ADOPTED SIGNAL PARAMETERS FOR THE DYNAMIC MEASUREMENTS OF THE ACOUSTIC POSITIONING SYSTEM

Parameter	Value
Sampling frequency f_s	32 kHz
Duration T	360 ms
Start frequency f_0	5 kHz
Stop frequency f_1	10 kHz

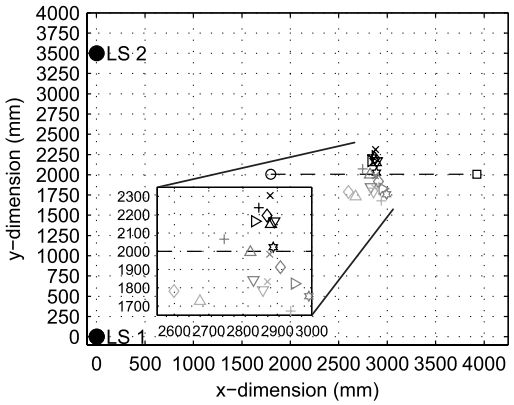


Fig. 15. Position estimation results from the DPT evaluation using LFM chirps with the velocities of the linear axis from Table V. The start and end positions of the linear axis are marked by a black circle and square, respectively.

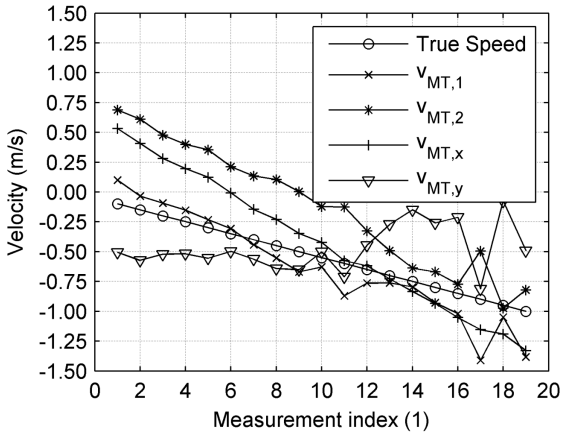


Fig. 16. Velocity estimation result from the DPT evaluation using LFM chirps.

consequential positioning plots. A photo of the setup is shown in Fig. 14. Furthermore, the signal parameters from Table II were adapted based on the experience gained by the static measurements and are summarized in Table IV. It has been experimentally determined that the correlation approach with LFM chirps fails for velocities > 0.5 m/s due to correlation signal distortion when the tag is in motion [31].

1) Measurements Using LFM Chirps and DPT Algorithm: As mentioned before, the DPT uses the DFT (9) for the spectral evaluation. To achieve a fine grid resolution, the DFT was performed with an equivalent zero padding of $Z = 2^{19}$. Moreover, a PE algorithm was applied to achieve an even higher precision. The position estimation results using LFM chirps and the DPT algorithm are shown in Fig. 15. Since the measurements were acquired during the movement of the axis, a reference position could not be determined. However, all measurements were started at the same point in time, and therefore, the position measurements should be located around the middle of the axis. Although each velocity resulted in a plausible position estimation, the precision of the overall measurements has evidentially decreased. Furthermore, the velocity estimates shown in Fig. 16 are also not very accurate and get worse as the speed increases. The $v_{\mathrm{MT},x}$ and $v_{\mathrm{MT},y}$ components are determined according to Appendix B.

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TABLE V
USED VELOCITIES AND THEIR MARKER DEFINITION FOR THE POSITION PLOTS. THE LAST TWO ENTRIES ARE NOT USED

(m/s)	×	+	◇	▽	△	▷	✡
black	0.10	0.15	0.20	0.25	0.30	0.35	0.40
gray	0.45	0.50	0.55	0.60	0.65	0.70	0.75
light gray	0.80	0.85	0.90	0.95	1.00	N/A	N/A

TABLE VI
ADOPTED SIGNAL PARAMETERS FOR THE DYNAMIC MEASUREMENTS OF THE ACOUSTIC POSITIONING SYSTEM

Parameter	LS 1	LS 2
Start frequency $f_{p,0}$	5 kHz	6 kHz
Stop frequency $f_{p,1}$	10 kHz	11 kHz

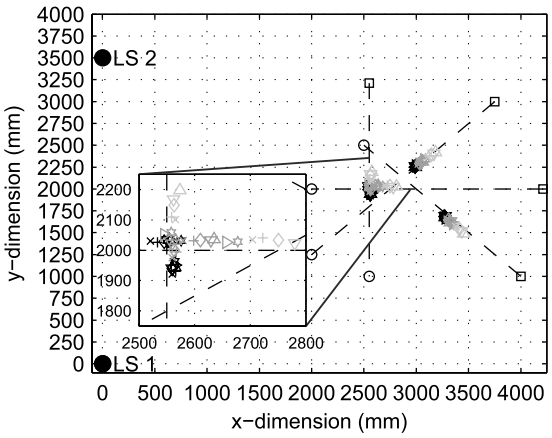


Fig. 17. Position estimates using HFM chirps for several alignments of the linear axis. The position estimation markers are set according to Table V. The start and end positions of the linear axis are marked by black squares and circles, respectively. The region of interest is zoomed in for clarity.

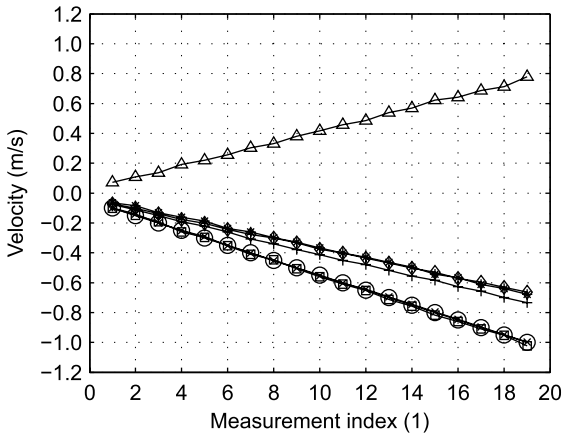


Fig. 18. Velocity estimates of two diagonal aligned linear axis from Fig. 17. In the following, the superscript D1 denotes the axis aligned from the lower left to the upper right point and D2, from the upper left to the lower right. reference velocity, $\hat{v}^{\mathrm{D1}}_{\mathrm{MT},x}$, $\hat{v}^{\mathrm{D1}}_{\mathrm{MT},y}$, $\hat{v}^{\mathrm{D2}}_{\mathrm{MT},x}$, $\hat{v}^{\mathrm{D2}}_{\mathrm{MT},y}$, $\|\hat{\mathbf{v}}^{\mathrm{D1}}_{\mathrm{MT}}\|$, and $\|\hat{\mathbf{v}}^{\mathrm{D2}}_{\mathrm{MT}}\|$. Note that the curves , , and , as well as the curves , , and almost overlap.

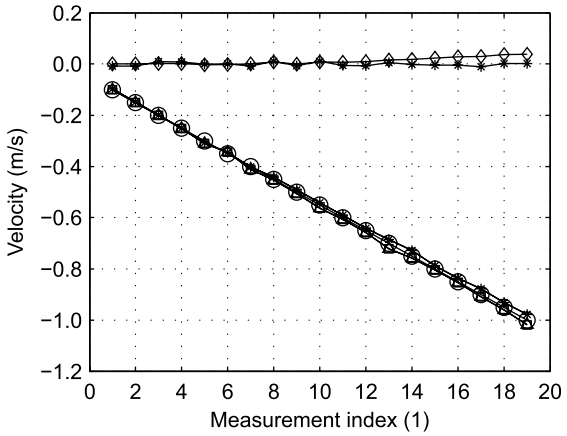


Fig. 19. Velocity estimates of the horizontal and vertical aligned linear axis from Fig. 17. In the following, the superscript H denotes the horizontal and V, the vertical aligned axis, respectively. reference velocity, $\hat{v}^{H}_{\mathrm{MT},x}$, $\hat{v}^{H}_{\mathrm{MT},y}$, $\hat{v}^{V}_{\mathrm{MT},x}$, $\hat{v}^{V}_{\mathrm{MT},y}$, $\|\hat{\mathbf{v}}^{H}_{\mathrm{MT}}\|$, and $\|\hat{\mathbf{v}}^{V}_{\mathrm{MT}}\|$. Note that the curves , , , , , and as well as the curves and almost overlap.

The plausible reason for the limited performance of the DPT may origin from lack of SNR in the first step where the second-order coefficient is estimated. A detailed evaluation revealed that the SNR can be down to 3 dB even for band-limited and filtered data. Since all measurements were acquired without any background noise, it is highly doubtable that the DPT approach will work in a noisy mine.

2) Measurements Using HFM Chirps and Correlation: To evaluate the TOF and velocity with HFM chirps, two chirp signal must be transmitted per LS. As previously stated, the HFM allows the frequency bands of two chirps to overlap and so the start and stop frequencies from Table VI were used. The other signal parameters are identical with Table IV. Again, the down-chirps are realized by exchanging $f_{p,0}$ with $f_{p,1}$ in Table VI.

The position estimation results using HFM chirps are shown in Fig. 17. Due to the good results, from the simple test from Fig. 14, the linear axis was repositioned several times to verify the performance. Clearly, all position estimation results are closely aligned along the movement direction of the linear axis and even the unknown acceleration of the linear axis becomes visible, since measurements with faster velocities are shifted toward the movement direction.

Also the velocity estimates shown in Figs. 18 and 19 are now in remarkable alignment with the reference velocities. For the sake of clarity, the graphs for the radial velocities are omitted and only the velocity vector components computed using Appendix B are shown. Furthermore, the L_2-norms of the velocity vectors are also plotted.

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V. Conclusion

We showed the application of an acoustic positioning system for a harsh underground environment. The pulse-compression technique with linear chirps is used as a basis for precise and accurate positioning even if the surrounding noise is louder than the measurement signals. Since the Doppler influence makes the received chirp signal dissimilar to the transmitted variant, the matched filter approach fails even for small radial velocities. Hence, we proposed either the application of the DPT to estimate the chirp parameters directly or hyperbolic chirp signals which are insensitive toward the Doppler influence. We have shown by measurements that the latter approach allows precise and accurate estimation of the position in the centimeter range as well as the velocity.

Remaining influences on the accuracy of the estimated positions such as multipath reflections and the uncertainty regarding the mounting of the LSs may be further reduced by the application of peak picking in the correlation signals and the calibration of the measurement system.

Appendix A
Using the DPT to Estimate TOF and Velocity of an LFM Chirp

Inserting the parameters from (19) into (21) and expansion yields

$$\phi^d_{\mathrm{LFM}}(t-\tau_p, v_{\mathrm{MT},p}) = \alpha\pi\nu_p t^2 + 2\pi\nu_p(f_0-\alpha\tau_p)t + \alpha\pi\nu_p\tau_p^2 - 2\pi\nu_p f_0\tau_p \tag{32}$$

with the abbreviation

$$\nu_p = 1 - \frac{v_{\mathrm{MT},p}}{c_s}. \tag{33}$$

Using the discrete-time equivalent of (32) results to

$$\phi^d_{\mathrm{LFM}}[n-\tau_p, v_{\mathrm{MT},p}] = a_2 n^2 + a_1 n + a_0 \tag{34}$$

where

$$a_2 = \beta\pi\nu_p \tag{35}$$

$$a_1 = 2\pi\nu_p(\psi_0 - \beta\tau_p) \tag{36}$$

$$a_0 = \beta\pi\nu_p\tau_p^2 - 2\pi\nu_p f_0\tau_p. \tag{37}$$

Using the DPT, (35)–(37) can be estimated and the Doppler contribution can be directly extracted from (35) by

$$\hat{\nu}_p = \frac{\hat{a}_2}{\pi\beta} \tag{38}$$

or with (33)

$$\hat{v}_{\mathrm{MT},p} = c_s\left(1 - \frac{\hat{a}_2}{\pi\beta}\right). \tag{39}$$

In the same vein, the TOF can be estimated inserting (38) in (36) and expressing τ_p to

$$\hat{\tau}_p = \frac{1}{\beta}\left(\psi_0 - \frac{\hat{a}_1}{2\pi\hat{\nu}_p}\right). \tag{40}$$

Appendix B
Conversion of Radial Velocity-to-Velocity Vector

As for the position estimation, at least P different LS positions must be available for estimating a P-dimensional velocity vector. If more measurements are available a least-squares approach has to be applied to solve the overdetermined system of equations. However, the following derivations will be performed only for the $P = 2$ case. The scalar radial projection of the velocity vector is given by

$$v_{\mathrm{MT},p} = \frac{\mathbf{r}_p \cdot \mathbf{v}_{\mathrm{MT}}}{\|\mathbf{r}_p\|_2}$$

with the directional vector toward the pth LS $\mathbf{r}_p = [\, r_{x,p}\ r_{y,p}\,]^T$ and the velocity vector $\mathbf{v}_{\mathrm{MT}} = [\, v_{\mathrm{MT},x}\ v_{\mathrm{MT},y}\,]^T$. Taking the vector norm to the left side and expanding the vectorial dot-product yields

$$\begin{bmatrix} v_{\mathrm{MT},1}\,\|\mathbf{r}_1\|_2 \\ v_{\mathrm{MT},2}\,\|\mathbf{r}_2\|_2 \end{bmatrix} = \begin{bmatrix} r_{x,1} & r_{y,1} \\ r_{x,2} & r_{y,2} \end{bmatrix}\begin{bmatrix} v_{\mathrm{MT},x} \\ v_{\mathrm{MT},y} \end{bmatrix} = \mathbf{A}\mathbf{v}_{\mathrm{MT}}. \tag{41}$$

By multiplying the left side with the inverse of $\mathbf{A}$, results to the desired velocity vector components

$$v_{\mathrm{MT},x} = \frac{r_{y,2}v_{\mathrm{MT},1}\|\mathbf{r}_1\|_2 - r_{y,1}v_{\mathrm{MT},2}\|\mathbf{r}_2\|_2}{r_{x,1}r_{y,2} - r_{x,2}r_{y,1}} \tag{42}$$

and

$$v_{\mathrm{MT},y} = \frac{r_{x,1}v_{\mathrm{MT},2}\|\mathbf{r}_2\|_2 - r_{x,2}v_{\mathrm{MT},1}\|\mathbf{r}_1\|_2}{r_{x,1}r_{y,2} - r_{x,2}r_{y,1}}. \tag{43}$$

Note that this methods fails if $\mathbf{A}$ is singular or, i.e., the MT and LS positions are aligned. However, this case is excluded by the restriction to the positive half-plane of the x-dimension.

Appendix C
Using an HFM Up-Chirp and Down-Chirp to Estimate TOF and Velocity

As mentioned before each LS transmits one up-chirp and down-chirp simultaneously, thus two time of arrivals (TOAs) can be estimated as

$$\hat{\tau}_{q,p} = T_s \arg\max\{s_{c,q}[n]\} \tag{44}$$

with $q = 1, 2$ and the correlation signal from the up-chirp $s_{c,1}$ and $s_{c,2}$ from the down-chirp, respectively. Based on (26), it can be concluded that the TOA is consists of the TOF and the Doppler contribution, or

$$\tau_{q,p} = \tau_p + \tau_{d,q,p}. \tag{45}$$

The Doppler term in (45) is dependent on q since the divisor of (27) is altered for an up-chirp and down-chirp. Hence, subtracting $q = 2$ from $q = 1$ yields an estimate for (28)

$$\hat{\phi}_p = (\tau_{2,p} - \tau_{1,p})\left(\frac{1}{b_{p,2}f_{p,2,0}} - \frac{1}{b_{p,1}f_{p,1,0}}\right)^{-1} \tag{46}$$

with the start frequency $f_{p,q,0}$ and the modulation index $b_{p,q}$ of the $q = 1$ up-chirp or $q = 2$ down-chirp, respectively,

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for the pth LS signal. Using (28) and (33) the velocity estimate from (46) can be derived to

$$\hat{v}_{\mathrm{MT},p} = c_s \left(1 - \frac{1}{1-\hat{\phi}_p}\right). \quad (47)$$

Again, (47) only describes the scalar projected velocity which has to be converted to $v_{\mathrm{MT},x}$ and $v_{\mathrm{MT},y}$ using (43) and (42), respectively. With (47) and the sum of (44) over q, the TOF can be estimated by

$$\hat{\tau}_p = \frac{1}{2}\left(\tau_{1,p} + \tau_{2,p} - \hat{\phi}_p \left(\frac{1}{b_{p,2} f_{p,2,0}} + \frac{1}{b_{p,1} f_{p,1,0}}\right)\right). \quad (48)$$

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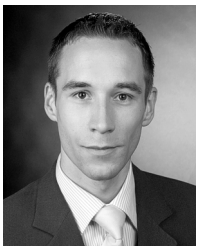
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